Bubble Sort

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The problem

- Repeatedly swap adjacent out-of-order pairs; stop early when a pass makes no swaps.

Intuition

- Repeatedly swap neighbours that are out of order; each pass bubbles the largest remaining item to the end.

Approach

- 1. Loop until a pass makes no swaps
- 2. Swap adjacent pairs that are out of order
- 3. Shrink the range each pass

Java solution

```java
public class BubbleSort {
    public static int[] bubbleSort(int[] a) {
        boolean swapped = true;
        for (int end = a.length - 1; swapped && end > 0; end--) {
            swapped = false;
            for (int i = 0; i < end; i++)
                if (a[i] > a[i + 1]) { int t = a[i]; a[i] = a[i + 1]; a[i + 1] = t; swapped = true; }
        }
        return a;
    }
}
```

Complexity

- Time: O(n^2) worst, O(n) best
- Space: O(1)

Recap

- Repeatedly swap neighbours that are out of order; each pass bubbles the largest remaining item to the end.
- Complexity: O(n^2) worst, O(n) best